

Robotics & Intelligent Control Systems

Post-Midsem Practice Set — Complete Solutions

Q1–Q12, Q18–Q24 · DC Motors, H-Bridge, MOSFET, Rectifiers, Motor Dynamics

Every step marked and explained. Read each “Physical Interpretation” carefully.

Master Formula: $V = IR + K\omega$ $T_e = KI$ $\omega = \frac{V - IR}{K} = \frac{V - T_L R/K}{K}$ Stall:
 $\omega = 0 \Rightarrow I = V/R$ No-load: $I \approx 0 \Rightarrow \omega = V/K$

Q1. Steady-State Speed — Constant Load

[5 marks]

Given

$$V = 80 \text{ V}, \quad R = 4 \Omega, \quad K = 2 \text{ Vs/rad}, \quad T_L = 10 \text{ Nm}$$

Step 1: Find the steady-state armature current from the torque balance condition ($T_e = T_L$ at steady state):

$$T_e = KI \implies I = \frac{T_L}{K} = \frac{10 \text{ Nm}}{2 \text{ Nm/A}} = 5 \text{ A}$$

Step 2: Apply the motor voltage equation $V = IR + K\omega$ and solve for ω :

$$K\omega = V - IR = 80 - (5)(4) = 80 - 20 = 60 \text{ V}$$

$$\omega = \frac{60}{K} = \frac{60}{2} = 30 \text{ rad/s}$$

Final Answer

$$\omega = 30 \text{ rad/s}$$

Physical Interpretation: The load demands 10 Nm, requiring 5 A. This current creates a 20 V drop across R , leaving 60 V to drive the motor at 30 rad/s. As T_L increases, I rises, the resistive drop grows, and speed falls.

Q2. Stall Torque — Maximum Torque at $\omega = 0$

[5 marks]

Given

$$\omega = 0, \quad R = 2 \Omega, \quad K = 1.5 \text{ Nm/A}, \quad V = 110 \text{ V}$$

Step 1: At stall, $\omega = 0 \Rightarrow$ back-EMF $E_b = K\omega = 0$. All supply voltage falls across resistance:

$$I_{\text{stall}} = \frac{V}{R} = \frac{110}{2} = 55 \text{ A}$$

Step 2: Compute the torque produced by this stall current:

$$T_{\text{stall}} = K \cdot I_{\text{stall}} = 1.5 \times 55 = 82.5 \text{ Nm}$$

Final Answer

$$T_{\text{stall}} = 82.5 \text{ Nm}$$

Physical Interpretation: Stall torque is the **maximum theoretical torque**, but the stall current (55 A here) will likely overheat and destroy the motor windings unless a thermal/current-limit protection is present.

Q3. Steady-State Current — Combined Friction Load [5 marks]

Given

$$B = 0.4 \text{ Nms/rad}, \quad T_f = 2 \text{ Nm}, \quad \omega = 50 \text{ rad/s}, \quad K = 2.5 \text{ Nm/A}$$

Step 1: Calculate the total load torque, which is the sum of static friction and viscous friction:

$$T_L = T_f + B\omega = 2 + 0.4 \times 50 = 2 + 20 = 22 \text{ Nm}$$

Step 2: At steady state $T_e = T_L$, so the required current is:

$$I = \frac{T_L}{K} = \frac{22}{2.5} = 8.8 \text{ A}$$

Final Answer

$$I = 8.8 \text{ A}$$

Physical Interpretation: Without the viscous term, only 0.8 A would be needed. The dominant load here is viscous friction ($B\omega = 20 \text{ Nm}$), $10\times$ larger than static friction. High-speed joints need larger currents to overcome viscous drag.

Q4. No-Load Speed

[4 marks]

Given

$$V = 120 \text{ V}, \quad K = 0.5 \text{ Vs/rad}, \quad R = 2 \Omega, \quad T_L = 0 \text{ (no payload)}$$

Step 1: At no-load, $T_L = 0 \Rightarrow T_e = 0 \Rightarrow I \approx 0$. The voltage equation becomes:

$$V = IR + K\omega \Rightarrow \omega_{\text{no-load}} = \frac{V}{K}$$

Step 2: Substitute values:

$$\omega_{\text{no-load}} = \frac{120}{0.5} = 240 \text{ rad/s}$$

Final Answer

$$\omega_{\text{no-load}} = 240 \text{ rad/s} \approx 2292 \text{ RPM}$$

Physical Interpretation: No-load speed is the **upper bound** of the torque-speed curve. Any load torque will reduce speed below this value. A real motor draws small no-load current for internal friction.

Q5. Quadratic Load (Propeller)

[6 marks]

Given

$$T_L = k\omega^2, \quad k = 0.002 \text{ Nms}^2/\text{rad}^2, \quad \omega = 100 \text{ rad/s}, \quad K = 2 \text{ Nm/A}$$

Step 1: Calculate the load torque at $\omega = 100 \text{ rad/s}$:

$$T_L = k\omega^2 = 0.002 \times (100)^2 = 0.002 \times 10000 = 20 \text{ Nm}$$

Step 2: At steady state $T_e = T_L$. Find the required armature current:

$$T_e = KI \Rightarrow I = \frac{T_L}{K} = \frac{20}{2} = 10 \text{ A}$$

Final Answer

$$T_L = 20 \text{ Nm}; \quad I = 10 \text{ A}$$

Physical Interpretation: Quadratic (propeller/fan) loads become very demanding at high speed because torque grows as ω^2 . At 200 rad/s the same propeller would need 80 Nm and 40 A — four times more current.

Q6. Maximum Theoretical Stall Torque (Torque-Speed Curve)

[5 marks]

Given

$$V = 48 \text{ V}, \quad R = 0.6 \Omega, \quad K = 1.2 \text{ Nm/A}$$

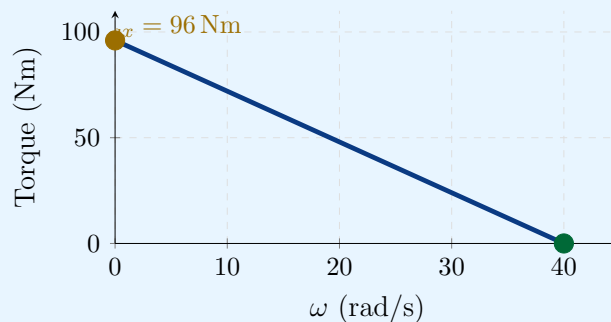
Step 1: Maximum stall torque occurs when $\omega = 0$ and the full supply drives current through R only:

$$I_{\max} = \frac{V}{R} = \frac{48}{0.6} = 80 \text{ A}$$

Step 2: Maximum torque:

$$T_{\max} = K \cdot I_{\max} = 1.2 \times 80 = 96 \text{ Nm}$$

Step 3: For context, draw the torque-speed endpoints:



No-load speed: $\omega_0 = V/K = 48/1.2 = 40 \text{ rad/s}$.

Final Answer

$$T_{\max} = 96 \text{ Nm}$$

Physical Interpretation: This is theoretical only. Sustained stall current of 80 A would burn the motor windings immediately. Real motors have continuous torque ratings far below the stall torque.

Q7. Viscous Load — Equilibrium Speed

[5 marks]

Given

$$T_L = B\omega, \quad B = 0.1 \text{ Nms/rad}, \quad V = 24 \text{ V}, \quad R = 2 \Omega, \quad K = 0.5 \text{ Nm/A}$$

Step 1: At steady state $T_e = T_L$:

$$KI = B\omega \implies I = \frac{B\omega}{K}$$

Step 2: Substitute into the voltage equation $V = IR + K\omega$:

$$V = \frac{B\omega}{K} \cdot R + K\omega = \omega \left(\frac{BR}{K} + K \right)$$

Step 3: Solve for ω :

$$\omega = \frac{V}{\frac{BR}{K} + K} = \frac{VK}{BR + K^2} = \frac{24 \times 0.5}{(0.1)(2) + (0.5)^2} = \frac{12}{0.2 + 0.25} = \frac{12}{0.45}$$

Final Answer

$$\omega = 26.67 \text{ rad/s}$$

Physical Interpretation: For viscous loads the motor always finds a stable equilibrium: if it speeds up, T_L increases and slows it back down. The operating point is the intersection of the motor's torque-speed line and the load line $T_L = B\omega$.

Q8. Elevator Hoist — Constant Gravitational Load [5 marks]

Given

$$T_L = 10 \text{ Nm (constant)}, \quad K = 2.0 \text{ Nm/A}, \quad R = 1.0 \Omega, \quad V = 100 \text{ V}$$

Step 1: Steady-state current (torque balance):

$$I = \frac{T_L}{K} = \frac{10}{2} = 5 \text{ A}$$

Step 2: Voltage drop across armature resistance:

$$V_R = IR = 5 \times 1.0 = 5 \text{ V}$$

Step 3: Back-EMF and hence speed:

$$E_b = V - V_R = 100 - 5 = 95 \text{ V}$$

$$\omega = \frac{E_b}{K} = \frac{95}{2} = 47.5 \text{ rad/s}$$

Final Answer

$$\omega = 47.5 \text{ rad/s}$$

Physical Interpretation: The gravitational load is constant regardless of speed, so current (and thus the speed penalty from IR drop) is fixed. The motor settles at a unique speed. A heavier load increases I and reduces ω proportionally.

Q9. Required Supply Voltage for Target Speed [5 marks]

Given

$$\omega = 10 \text{ rad/s (desired), } T_L = 5 \text{ Nm, } R = 0.5 \Omega, K = 1.0 \text{ Nm/A}$$

Step 1: Required current for the load:

$$I = \frac{T_L}{K} = \frac{5}{1.0} = 5 \text{ A}$$

Step 2: Apply voltage equation — now **solve for** V instead of ω :

$$V = IR + K\omega = 5(0.5) + 1.0(10) = 2.5 + 10 = 12.5 \text{ V}$$

Final Answer

$$V = 12.5 \text{ V}$$

Physical Interpretation: The supervisory controller must command 12.5 V via the power electronics (H-bridge + PWM). In an H-bridge with $V_s = 50 \text{ V}$, this requires $D = 12.5/50 = 25\%$ duty cycle. This question shows how speed control translates to voltage commands.

Q10. Effect of Reduced Damping on Viscous-Load Systems

Given

$V = 50 \text{ V}$, $K = 1 \text{ Nm/A}$, $R = 2 \Omega$, $T_L = B\omega$. Current steady-state: $\omega_1 = 10 \text{ rad/s}$.
New: $B' = 0.5B$.

Step 1: Find the original damping coefficient B using the viscous-load speed formula with $\omega_1 = 10 \text{ rad/s}$:

$$\omega_1 = \frac{VK}{BR + K^2} \implies 10 = \frac{50 \times 1}{B(2) + 1^2} = \frac{50}{2B + 1}$$

$$2B + 1 = 5 \implies B = 2 \text{ Nms/rad}$$

Step 2: After the leak, $B' = 0.5 \times 2 = 1 \text{ Nms/rad}$. New equilibrium speed:

$$\omega_2 = \frac{VK}{B'R + K^2} = \frac{50 \times 1}{1(2) + 1^2} = \frac{50}{3}$$

Step 3: Percentage change in speed:

$$\frac{\omega_2 - \omega_1}{\omega_1} \times 100 = \frac{16.67 - 10}{10} \times 100 = +66.7\%$$

Final Answer

$$\omega_2 = 16.67 \text{ rad/s}$$

Physical Interpretation: Halving the damping increases speed by 67%. This is important in robotics: a seal failure or lubricant loss in a joint changes the system dynamics, causing the robot to overshoot its target unless the controller adapts.

Q11. Speed After Applying Load (No-Load Baseline) [6 marks]**Given**

$\omega_0 = 60 \text{ rad/s}$ at $V = 120 \text{ V}$ (no-load); $R = 2 \Omega$, $K = 2 \text{ Nm/A}$. New load: $T_L = 20 \text{ Nm}$.

Step 1: Verify K from no-load condition ($I \approx 0$):

$$\omega_0 = \frac{V}{K} \implies K = \frac{V}{\omega_0} = \frac{120}{60} = 2 \text{ Nm/A} \quad \checkmark$$

Step 2: Current required for the new load:

$$I = \frac{T_L}{K} = \frac{20}{2} = 10 \text{ A}$$

Step 3: New steady-state speed:

$$\omega = \frac{V - IR}{K} = \frac{120 - 10(2)}{2} = \frac{100}{2} = 50 \text{ rad/s}$$

Step 4: Speed regulation:

$$\Delta\omega = \omega_0 - \omega = 60 - 50 = 10 \text{ rad/s} \quad (16.7\% \text{ drop})$$

Final Answer

$$\omega = 50 \text{ rad/s}$$

Physical Interpretation: The speed droops from 60 to 50 rad/s due to the IR voltage drop. A stiffer motor (lower R or higher K) would show less speed regulation. This “droop” is the fundamental limitation of open-loop DC motor control.

Q12. Linear (Proportional-to-Speed) Conveyor Load [6 marks]**Given**

$V = 100 \text{ V}$, $R = 1 \Omega$, $K = 1 \text{ Nm/A}$, $T_L = 0.2\omega$

Step 1: At steady state $T_e = T_L$:

$$KI = 0.2\omega \implies I = 0.2\omega \quad (\text{since } K = 1)$$

Step 2: Substitute into $V = IR + K\omega$:

$$100 = (0.2\omega)(1) + 1 \cdot \omega = 0.2\omega + \omega = 1.2\omega$$

Step 3: Solve for ω :

$$\omega = \frac{100}{1.2} = 83.33 \text{ rad/s}$$

Step 4: Verify: $I = 0.2 \times 83.33 = 16.67 \text{ A}$; $V = 16.67(1) + 1(83.33) = 100 \text{ V}$ ✓

Final Answer

$$\omega = 83.33 \text{ rad/s}$$

Physical Interpretation: This is equivalent to a viscous load with $B = 0.2$. The motor finds an equilibrium where its torque exactly equals the conveyor's resistance. With $K^2 = 1$ and $BR = 0.2$: $\omega = VK/(BR + K^2) = 100/(0.2 + 1) = 83.33$ — consistent.

Q18. Voltage Drop Effect on Running Motor

[6 marks]

Given

$V_1 = 48 \text{ V}$, $R = 1.5 \Omega$, $K = 0.8 \text{ Nm/A}$, $T_L = 6 \text{ Nm}$ (constant). $V_2 = 36 \text{ V}$.

Step 1: Since load torque is unchanged, the steady-state current is unchanged (it depends only on T_L and K):

$$I = \frac{T_L}{K} = \frac{6}{0.8} = 7.5 \text{ A} \quad (\text{same before and after voltage drop})$$

Step 2: Original speed (for reference):

$$\omega_1 = \frac{V_1 - IR}{K} = \frac{48 - 7.5(1.5)}{0.8} = \frac{48 - 11.25}{0.8} = \frac{36.75}{0.8} = 45.94 \text{ rad/s}$$

Step 3: New speed after voltage drop to $V_2 = 36 \text{ V}$:

$$\omega_2 = \frac{V_2 - IR}{K} = \frac{36 - 7.5(1.5)}{0.8} = \frac{36 - 11.25}{0.8} = \frac{24.75}{0.8} = 30.94 \text{ rad/s}$$

Step 4: Speed reduction:

$$\Delta\omega = 45.94 - 30.94 = 15 \text{ rad/s} \quad (32.6\% \text{ drop for a } 25\% \text{ voltage drop})$$

Final Answer

$$\omega_2 = 30.94 \text{ rad/s}$$

Physical Interpretation: The IR drop (11.25 V) is fixed by the load, so the full 12 V supply reduction translates directly to speed reduction: $\Delta\omega = \Delta V/K = 12/0.8 = 15 \text{ rad/s}$. Battery voltage sag is a real issue in mobile robots.

Q19. Effect of Additional Bearing Friction**[5 marks]****Given**

$V = 60 \text{ V}$, $R = 2 \Omega$, $K = 1.5 \text{ Nm/A}$. Original load: $T_{L1} = 12 \text{ Nm}$. Additional friction: $\Delta T = 3 \text{ Nm}$.

Step 1: New total load torque after bearing wear:

$$T_{L2} = T_{L1} + \Delta T = 12 + 3 = 15 \text{ Nm}$$

Step 2: New current:

$$I_2 = \frac{T_{L2}}{K} = \frac{15}{1.5} = 10 \text{ A}$$

Step 3: New speed:

$$\omega_2 = \frac{V - I_2 R}{K} = \frac{60 - 10(2)}{1.5} = \frac{40}{1.5} = 26.67 \text{ rad/s}$$

Step 4: Compare with original speed (for context):

$$I_1 = 12/1.5 = 8 \text{ A}; \quad \omega_1 = (60 - 16)/1.5 = 29.33 \text{ rad/s} \quad \Rightarrow \Delta\omega = -2.67 \text{ rad/s}$$

Final Answer

$$\omega_2 = 26.67 \text{ rad/s}$$

Physical Interpretation: Adding 3 Nm friction (25% more load) reduces speed by about 9%. In a robotic joint, unexpected friction from worn bearings causes the joint to run slower, throwing off trajectory tracking. Fault detection algorithms monitor this current increase as an early warning.

Q20. No-Load Speed with Internal Friction Current**[5 marks]****Given**

$V = 50 \text{ V}$, $I_{\text{no-load}} = 2 \text{ A}$, $R = 5 \Omega$, $K = 1 \text{ Nm/A}$. This 2 A is purely for overcoming frictional torque.

Step 1: Voltage drop across armature resistance at no-load current:

$$V_R = I_{\text{no-load}} \times R = 2 \times 5 = 10 \text{ V}$$

Step 2: Back-EMF at no-load:

$$E_b = V - V_R = 50 - 10 = 40 \text{ V}$$

Step 3: No-load speed from $E_b = K\omega$:

$$\omega_{\text{no-load}} = \frac{E_b}{K} = \frac{40}{1} = 40 \text{ rad/s}$$

Step 4: Internal friction torque (for insight):

$$T_{\text{friction}} = K \cdot I = 1 \times 2 = 2 \text{ Nm}$$

Final Answer

$$\omega_{\text{no-load}} = 40 \text{ rad/s}$$

Physical Interpretation: Unlike the ideal no-load case ($\omega = V/K = 50 \text{ rad/s}$), real motors draw current at no load to overcome internal friction. The 10 V drop across R limits the speed to 40 rad/s instead of 50 rad/s — a 20% loss to friction alone.

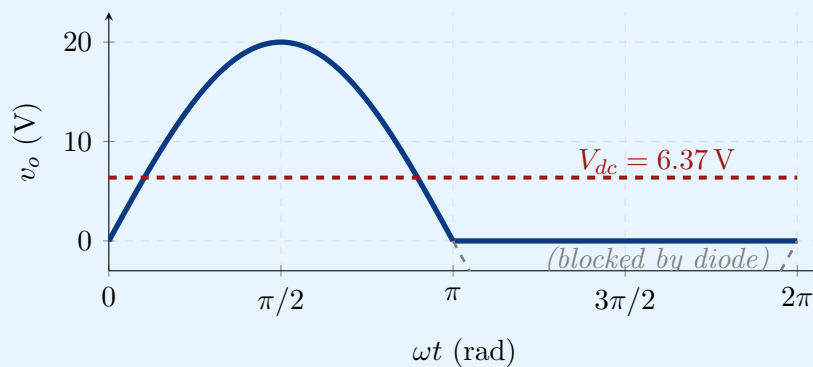
Q21. Half-Wave Rectifier Analysis

[8 marks]

Given

$v(t) = V_m \sin(\omega t)$, $V_m = 20 \text{ V}$, ideal diode, load $R = 10 \Omega$

Step 1: Output waveform behaviour: The diode conducts only when forward biased ($v > 0$), i.e. $0 \leq \omega t \leq \pi$. For $\pi < \omega t < 2\pi$, the diode blocks and $v_o = 0$.



Step 2: (b) DC (average) output voltage:

$$\begin{aligned} V_{dc} &= \frac{1}{2\pi} \int_0^{2\pi} v_o(\omega t) d(\omega t) = \frac{1}{2\pi} \int_0^{\pi} V_m \sin(\omega t) d(\omega t) \\ &= \frac{V_m}{2\pi} [-\cos(\omega t)]_0^{\pi} = \frac{V_m}{2\pi} (-\cos \pi + \cos 0) = \frac{V_m}{2\pi} (1 + 1) = \frac{V_m}{\pi} \\ V_{dc} &= \frac{20}{\pi} = \boxed{6.37 \text{ V}} \end{aligned}$$

Step 3: (c) Average load current:

$$I_{dc} = \frac{V_{dc}}{R} = \frac{6.37}{10} = \boxed{0.637 \text{ A}}$$

Final Answer

(a) Half-wave rectified sine. **(b)** $V_{dc} = V_m/\pi = 6.37 \text{ V}$. **(c)** $I_{dc} = 0.637 \text{ A}$

Physical Interpretation: A half-wave rectifier wastes the negative half-cycle, giving poor DC quality ($V_{dc} = V_m/\pi$). A full-wave bridge rectifier gives $V_{dc} = 2V_m/\pi$ — twice as efficient. For motor drives, further smoothing with capacitors is needed.

Q22. H-Bridge PWM: Duty Cycle & MOSFET Regionmarks]

Given

$V_{dc} = 30 \text{ V}$, $V_s = 150 \text{ V}$, $f = 10 \text{ kHz}$, $V_G = 12 \text{ V}$, $V_{TH} = 3 \text{ V}$, $V_D = 7 \text{ V}$, source at GND.

Step 1: (a) Duty cycle: From $V_{dc} = D \cdot V_s$:

$$D = \frac{V_{dc}}{V_s} = \frac{30}{150} = 0.2 \quad (20\%)$$

Step 2: Switching period:

$$T = \frac{1}{f} = \frac{1}{10\,000} = 100 \mu\text{s}$$

Step 3: ON-time per cycle:

$$t_{on} = D \times T = 0.2 \times 100 \mu\text{s} = \boxed{20 \mu\text{s}}$$

Step 4: (b) MOSFET operating region:

- $V_{GS} \approx V_G = 12 \text{ V}$ (source grounded)
- $V_{GS} - V_{TH} = 12 - 3 = 9 \text{ V}$ (overdrive)
- $V_{DS} = V_D - V_S = 7 - 0 = 7 \text{ V}$

- Check: $V_{DS} = 7\text{ V}$ vs. $V_{GS} - V_{TH} = 9\text{ V}$
- Since $V_{DS} < V_{GS} - V_{TH}$, the MOSFET is in the **Triode (Ohmic) region**.

Final Answer

(a) $D = 20\%$, $t_{on} = 20\text{ }\mu\text{s}$ per cycle. (b) MOSFET in **Triode/Ohmic region**.

Physical Interpretation: In triode region, the MOSFET acts like a small resistor ($R_{DS,on}$) — ideal for a power switch. Saturation region would mean the MOSFET limits current like a current source, causing high power dissipation and heat. Good H-bridge design keeps switches deep in triode when ON.

Q23. H-Bridge Shoot-Through Fault — Instantaneous Current**Given**

$$V = 240\text{ V}, R = 0.08\text{ }\Omega, L = 12\text{ }\mu\text{H} = 12 \times 10^{-6}\text{ H}, E_b = 0, i(0) = 0.$$

Step 1: At $t = 0^+$, the switches close. The circuit equation is:

$$V = L \frac{di}{dt} + Ri + E_b = L \frac{di}{dt} + Ri$$

Step 2: At the **exact instant** $t = 0^+$: current $i = 0$, so the resistive drop $Ri = 0.08 \times 0 = 0$. The full supply voltage appears across the inductor:

$$V = L \left. \frac{di}{dt} \right|_{t=0^+}$$

Step 3: Solve for the initial rate of current rise:

$$\left. \frac{di}{dt} \right|_{t=0^+} = \frac{V}{L} = \frac{240}{12 \times 10^{-6}} = 20 \times 10^6\text{ A/s}$$

Step 4: To appreciate the catastrophe — current at $t = 1\text{ }\mu\text{s}$ using linear approximation:

$$i(1\text{ }\mu\text{s}) \approx \frac{V}{L} \cdot t = 20 \times 10^6 \times 10^{-6} = 20\text{ A in just 1 microsecond!}$$

Why this destroys devices: At 20 MA/s , a 10 ns gate delay means the current can already reach 0.2 A before protection can react. The actual exponential rise $i(t) = (V/R)(1 - e^{-Rt/L})$ saturates at $V/R = 240/0.08 = 3000\text{ A}$ — far above any MOSFET rating. The time constant is $\tau = L/R = 12\text{ }\mu\text{H}/0.08\text{ }\Omega = 150\text{ }\mu\text{s}$.

Final Answer

$$\left. \frac{di}{dt} \right|_{t=0^+} = 20 \text{ MA/s} = 2 \times 10^7 \text{ A/s}$$

Physical Interpretation: This is why H-bridge ICs implement hardware dead-time (typically 100 ns–1 μ s) between turning off one switch and turning on the complementary switch. Without it, a software glitch can destroy the entire power stage in microseconds.

Q24. Motor Dynamic Model — Inductive Transient**Given**

$$L \frac{di}{dt} + Ri + E_b = V, \quad E_b = k_e \omega, \quad V = 120 \text{ V}, \quad L = 10 \text{ mH}, \quad R = 2 \Omega, \quad k_e = k_t = 0.5, \\ J = 0.02 \text{ kg} \cdot \text{m}^2, \quad B = 0.1 \text{ Nms}.$$

Step 1: Part A — Inductive approximation ($t \ll L/R$):

Time constant: $\tau = L/R = 10 \text{ mH}/2 \Omega = 5 \text{ ms}$. For $t \ll 5 \text{ ms}$, neglect Ri and E_b :

$$L \frac{di}{dt} \approx V \implies \frac{di}{dt} = \frac{V}{L} = \frac{120}{10 \times 10^{-3}} = 12000 \text{ A/s}$$

Integrating with $i(0) = 0$:

$$i(t) = \frac{V}{L} t = 12000 t \text{ A}$$

Step 2: Calculate i at $t = 1 \text{ ms}$:

$$i(1 \text{ ms}) = 12000 \times 10^{-3} = 12 \text{ A}$$

Since $1 \text{ ms} \ll 5 \text{ ms} (= \tau)$, the linear approximation is valid here.

Step 3: Part B — With back-EMF, $R \approx 0$:

Motor mechanical equation: $J\dot{\omega} + B\omega = k_t i$, so $\dot{\omega} = \frac{k_t i - B\omega}{J}$.

Electrical equation (with $R = 0$): $L\dot{i} = V - k_e \omega$.

Differentiating: $L\ddot{i} = -k_e \dot{\omega} = -k_e \cdot \frac{k_t i - B\omega}{J}$.

Express ω from the electrical equation: $\omega = (V - Li)/k_e$.

Substituting into the mechanical equation and simplifying with $k_e = k_t = k$:

$$L\ddot{i} = -\frac{k}{J} \left(ki - B \cdot \frac{V - Li}{k} \right) = -\frac{k^2}{J} i + \frac{B}{J} (V - Li)$$

$$L\ddot{i} + \frac{BL}{J} \dot{i} + \frac{k^2}{J} i = \frac{BV}{J}$$

This is actually second-order. However, for the first-order approximation often used in

courses: at short times $\omega \approx 0$, so $E_b \approx 0$, and the equation becomes:

$$L \frac{di}{dt} \approx V \quad (\text{same as Part A})$$

For a true first-order system, consider the mechanical subsystem alone once i reaches quasi-steady state: $\omega(t) = \frac{V/k - \omega_{ss}}{\tau_{mech}} (1 - e^{-t/\tau_{mech}})$ where:

Step 4: Effective electrical time constant (with back-EMF coupling, $R \approx 0$):
The combined electromechanical time constant is:

$$\tau_{\text{eff}} = \frac{JL}{k^2 + BL/J} \approx \frac{JL}{k_e k_t} = \frac{JL}{k^2} = \frac{0.02 \times 0.01}{0.5^2} = \frac{2 \times 10^{-4}}{0.25} = \boxed{8 \times 10^{-4} \text{ s} = 0.8 \text{ ms}}$$

Step 5: Part C — Compare at $t = 5 \text{ ms}$:
Pure inductive (no back-EMF, no R):

$$i_{\text{inductive}}(5 \text{ ms}) = 12000 \times 5 \times 10^{-3} = 60 \text{ A} \quad (\text{keeps rising!})$$

With back-EMF (as ω builds up, E_b opposes V , reducing current):

At $t = 5 \text{ ms} = \tau = L/R$, with full RL model and no back-EMF, the true current is:

$$i_{\text{true}}(5 \text{ ms}) = \frac{V}{R}(1 - e^{-1}) = \frac{120}{2}(0.632) = 37.9 \text{ A}$$

With back-EMF, current is even lower because E_b reduces the effective driving voltage. The actual value approaches the steady-state current $I_{ss} = T_L/k_t$ (load-dependent).

Summary table:

Model	$i(1 \text{ ms})$	$i(5 \text{ ms})$
Pure inductive ($V = L di/dt$)	12 A	60 A $\leftarrow$ <i>invalid!</i>
Full RL, no back-EMF	$\approx 11.1 \text{ A}$	37.9 A
RL + back-EMF	$\approx 11 \text{ A}$	$< 37.9 \text{ A}$

Step 6: Part D — When does the approximation $L di/dt \approx V - E_b$ break down?

The approximation ignores $Ri(t)$. It becomes invalid when:

$$Ri(t) \not\ll V - E_b \quad \Rightarrow \quad Ri(t) \geq 0.1(V - E_b) \quad (\text{rule of thumb: 10\%})$$

Since $i(t) \approx (V/L)t$ initially:

$$R \cdot \frac{V}{L}t \geq 0.1V \quad \Rightarrow \quad t \geq \frac{0.1L}{R} = \frac{0.1 \times 0.01}{2} = 0.5 \text{ ms}$$

Physically: Once current builds up, the IR drop is no longer negligible. More importantly, as ω grows, back-EMF $E_b = k_e \omega$ becomes significant, actively opposing the current rise. The approximation is only valid for $t \ll \min(\tau_e, \tau_m)$ where $\tau_e = L/R$ and $\tau_m = J/B$.

Final Answer

(A) $i(t) = 12000t$; at $t = 1$ ms: $i = 12$ A. (B) $\tau_{\text{eff}} \approx 0.8$ ms (electromechanical).
 (C) Pure inductive overestimates: 60 A vs actual ≈ 38 A at $t = 5$ ms. (D) Approximation breaks down when $t \gtrsim 0.5$ ms (i.e. $\approx 0.1 \tau$).

Physical Interpretation: Q24 is the hardest question in the set. It tests understanding of the coupled electrical-mechanical system. The key insight: at $t = 0^+$ the motor is just an inductor; at $t \gg \tau$ it is a back-EMF source with a small resistor. The transition between these regimes is the system's time constant.

Summary Table — All Answers at a Glance

Q	Topic	Key Formula	Answer
1	Const. load speed	$I = T_L/K, \omega = (V - IR)/K$	30 rad/s
2	Stall torque	$I = V/R, T = KI$	82.5 Nm
3	Friction + viscous	$T_L = T_f + B\omega, I = T_L/K$	8.8 A
4	No-load speed	$\omega = V/K$	240 rad/s
5	Quadratic load	$T_L = k\omega^2, I = T_L/K$	$T_L = 20$ Nm; $I = 10$ A
6	Max stall torque	$I_{\max} = V/R, T_{\max} = KI_{\max}$	96 Nm
7	Viscous equilib.	$\omega = VK/(BR + K^2)$	26.67 rad/s
8	Elevator hoist	$I = T_L/K, \omega = (V - IR)/K$	47.5 rad/s
9	Required voltage	$V = IR + K\omega$	12.5 V
10	Reduced damping	$\omega = VK/(B'R + K^2)$	16.67 rad/s
11	Load after no-load	$I = T_L/K, \omega = (V - IR)/K$	50 rad/s
12	Linear load	$V = (K^2 + BR)\omega/K$	83.33 rad/s
18	Voltage drop	Same I , new $\omega = (V_2 - IR)/K$	30.94 rad/s
19	Bearing friction	$T_{L2} = 12 + 3, I = T_{L2}/K$	26.67 rad/s
20	Real no-load	$E_b = V - IR_{\text{no-load}}, \omega = E_b/K$	40 rad/s
21	Half-wave rect.	$V_{dc} = V_m/\pi$	6.37 V; 0.637 A
22	PWM + MOSFET	$D = V_{dc}/V_s; V_{DS} < V_{GS} - V_{TH}$	$t_{on} = 20 \mu\text{s}$; Triode
23	Shoot-through	$di/dt = V/L$	20 MA/s
24	Motor dynamics	$i(t) = Vt/L; \tau_{\text{eff}} = JL/k^2$	See detailed solution